Practical 9

Data Visualization II

[ ]:
```
## Title Data Visualization II
```

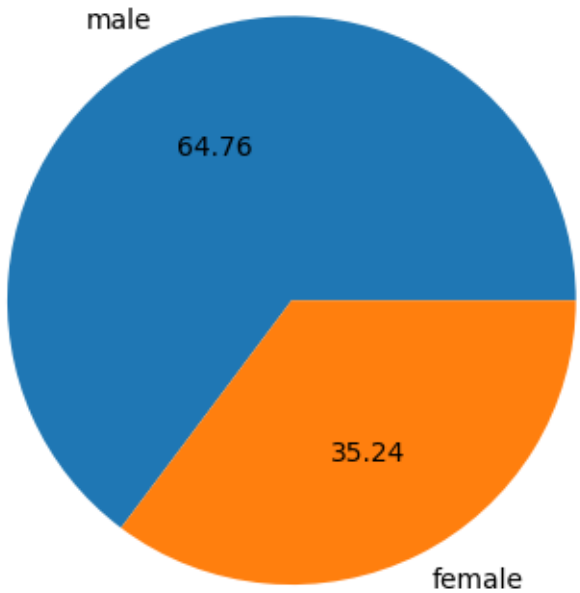
[40]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from seaborn import load_dataset

#titanic dataset
data = pd.read_csv("titanic_train.csv")

#tips dataset
tips = sns.load_dataset("tips")
```

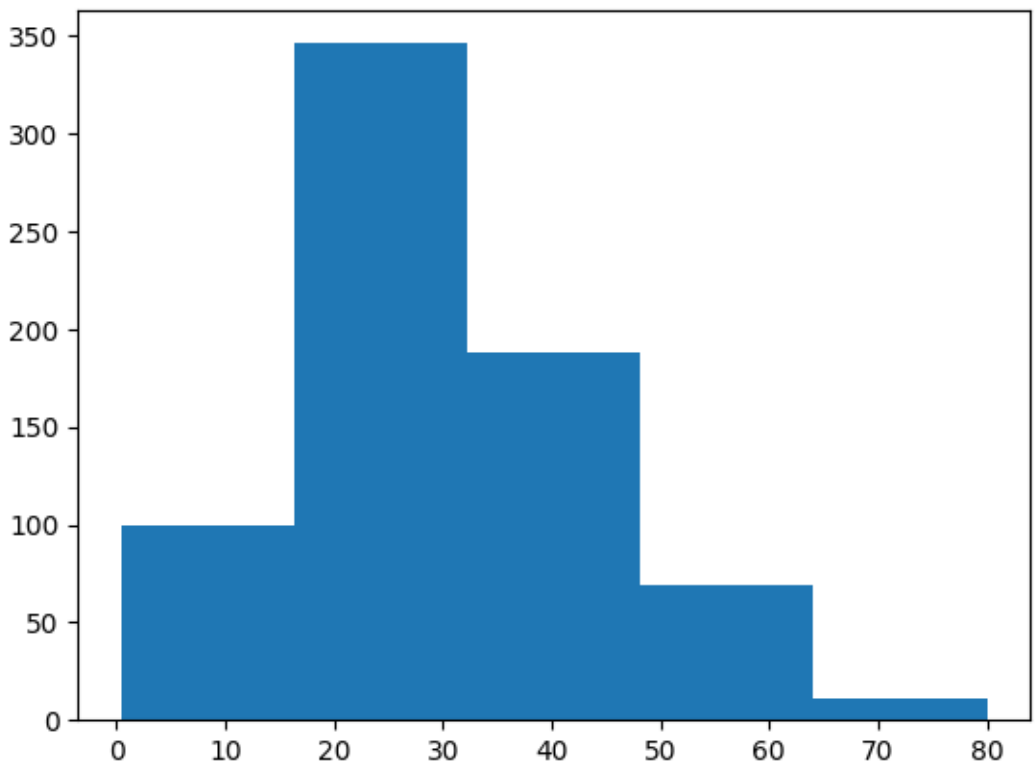
[41]:
```
## Pie Chart of Sex
```

[42]:
```
data['Sex'].value_counts().plot(kind="pie", autopct="%.2f")
plt.show()
```



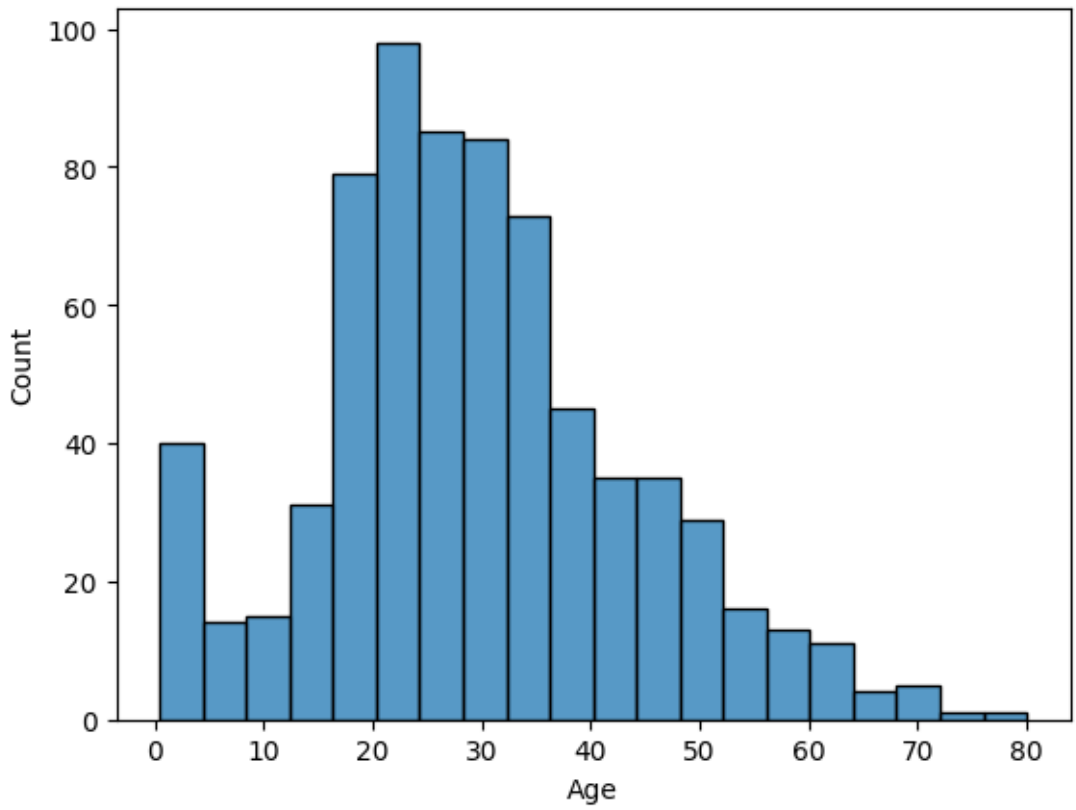
[43]:
```
## Histogram of Age
```

[44]:
```
plt.hist(data['Age'], bins=5)
plt.show()
```



[45]:
```
## Distplot of Age
```

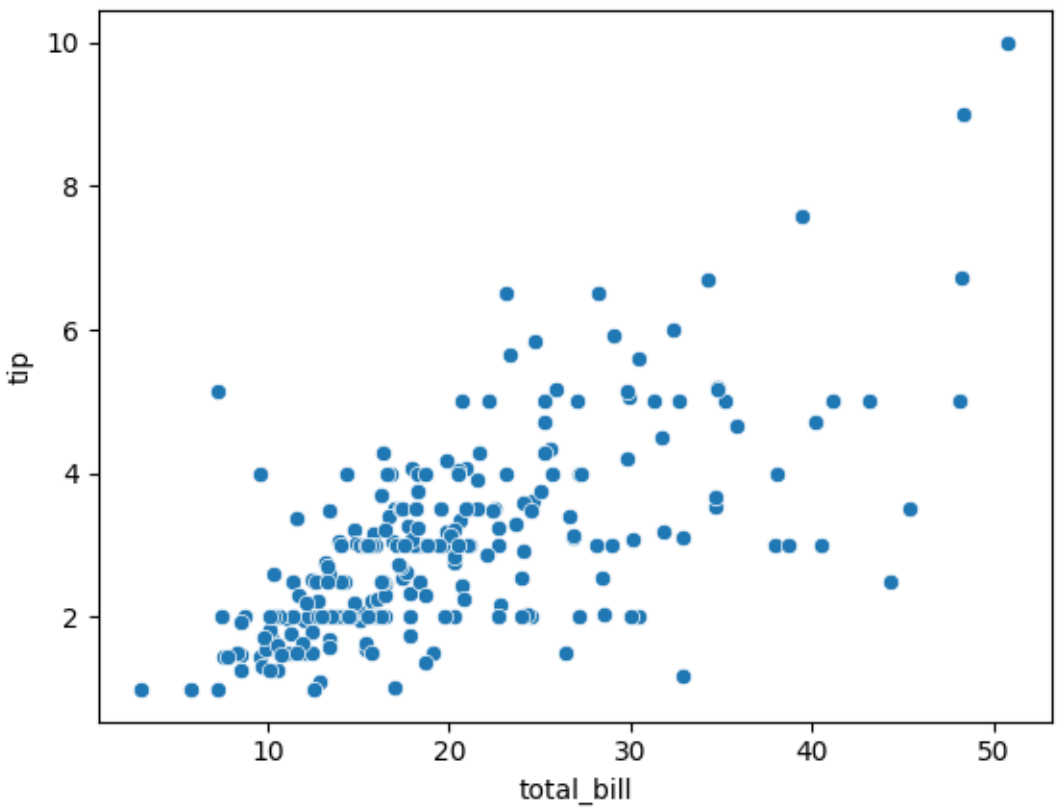
[46]:
```
sns.histplot(data['Age'])
plt.show()
```



[47]:
```
## Scatter Plot of Total Bill vs Tip
```

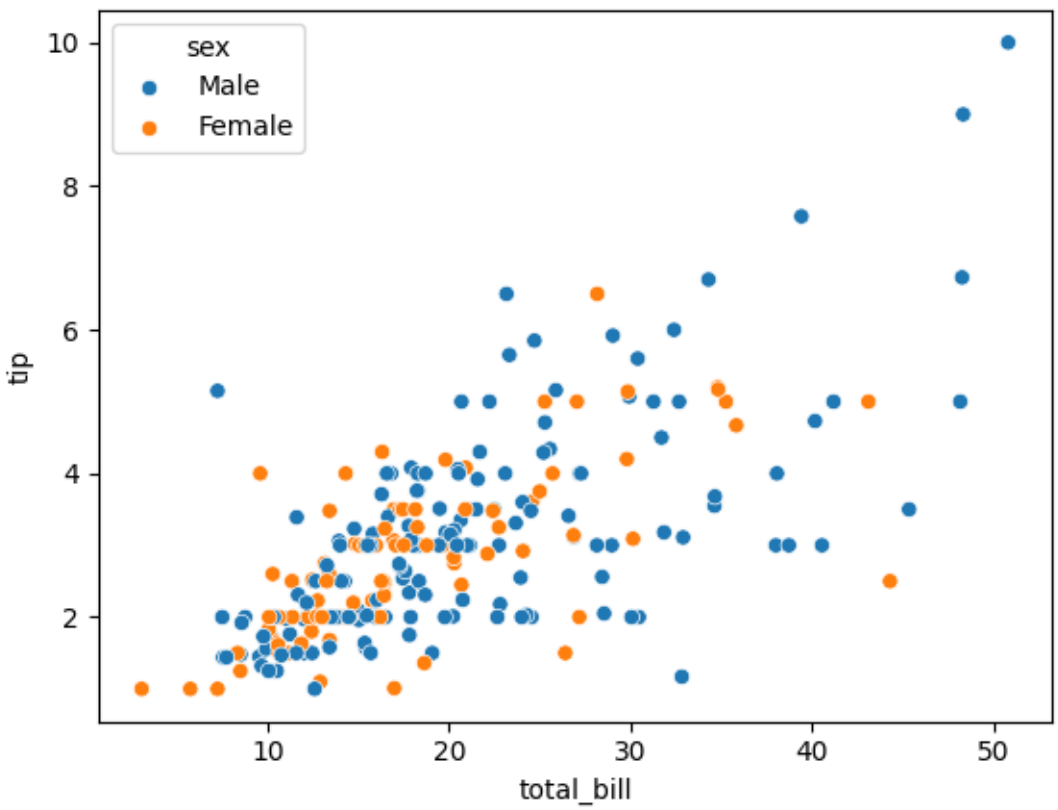
[48]:
```
sns.scatterplot(x="total_bill", y="tip", data=tips)
```

[48]:
```
<Axes: xlabel='total_bill', ylabel='tip'>
```



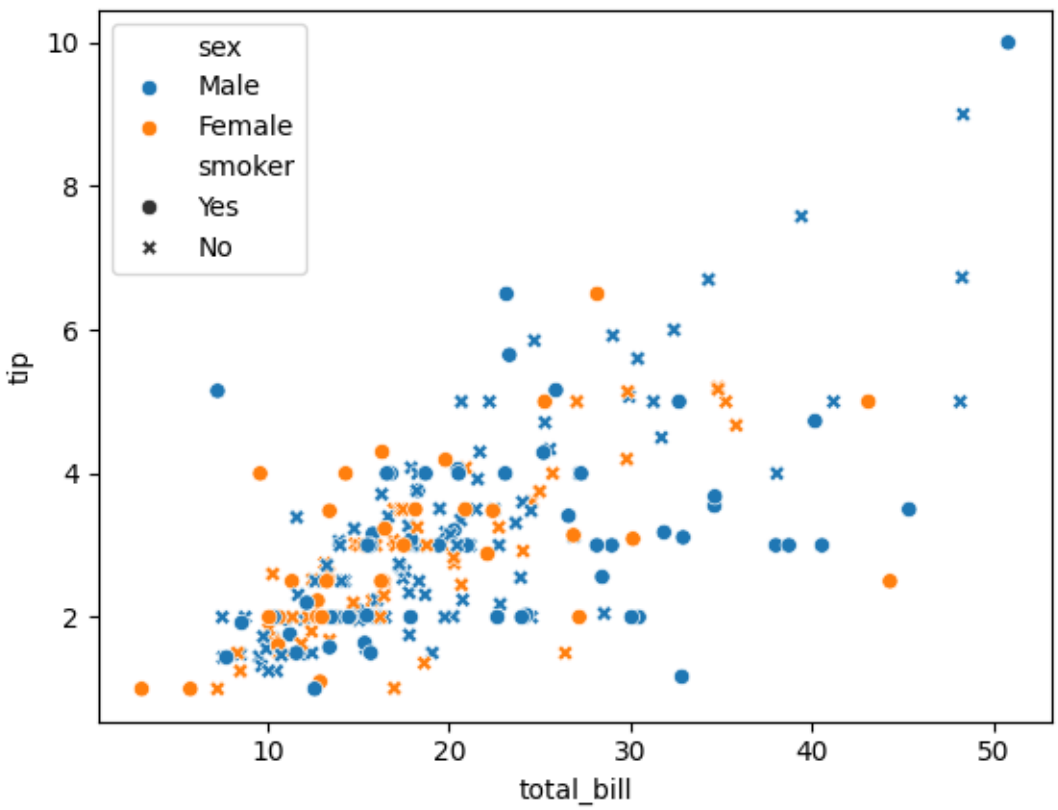
[49]:
```
## Multivariate Scatter Plot (Total Bill, Tip, Sex)
```

[50]:
```
sns.scatterplot(x="total_bill", y="tip", hue="sex", data=tips)
plt.show()
```



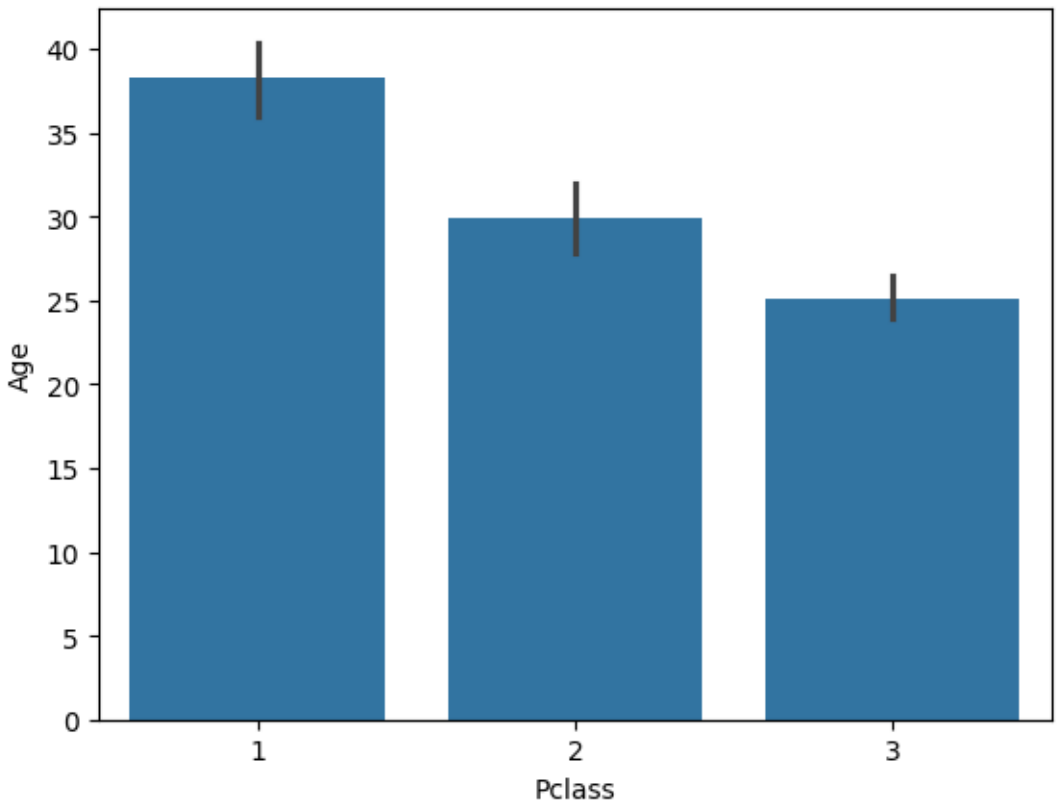
[51]:
```
## Multivariate Scatter Plot (Total Bill, Tip, Sex, Smoker)
```

[52]:
```
sns.scatterplot(x="total_bill", y="tip", hue="sex", style="smoker", data=tips)
plt.show()
```



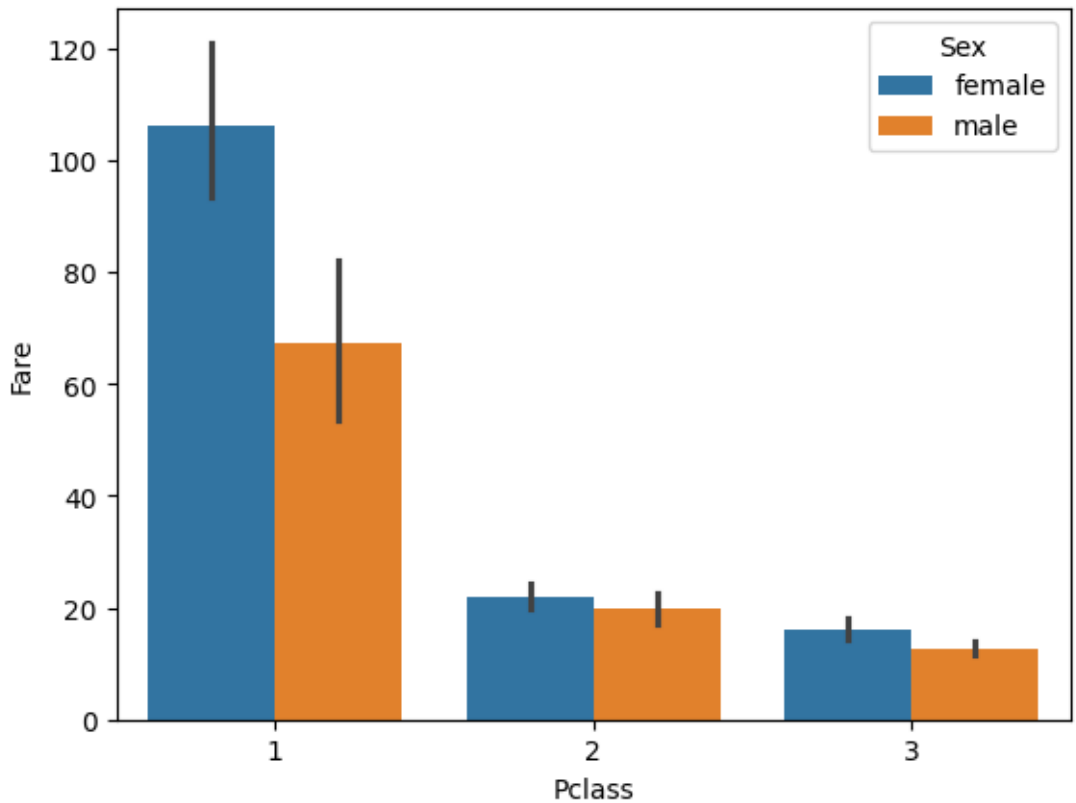
[53]:
```
## Bar Plot of Pclass vs Age
```

[54]:
```
sns.barplot(x="Pclass", y="Age", data=data)
plt.show()
```



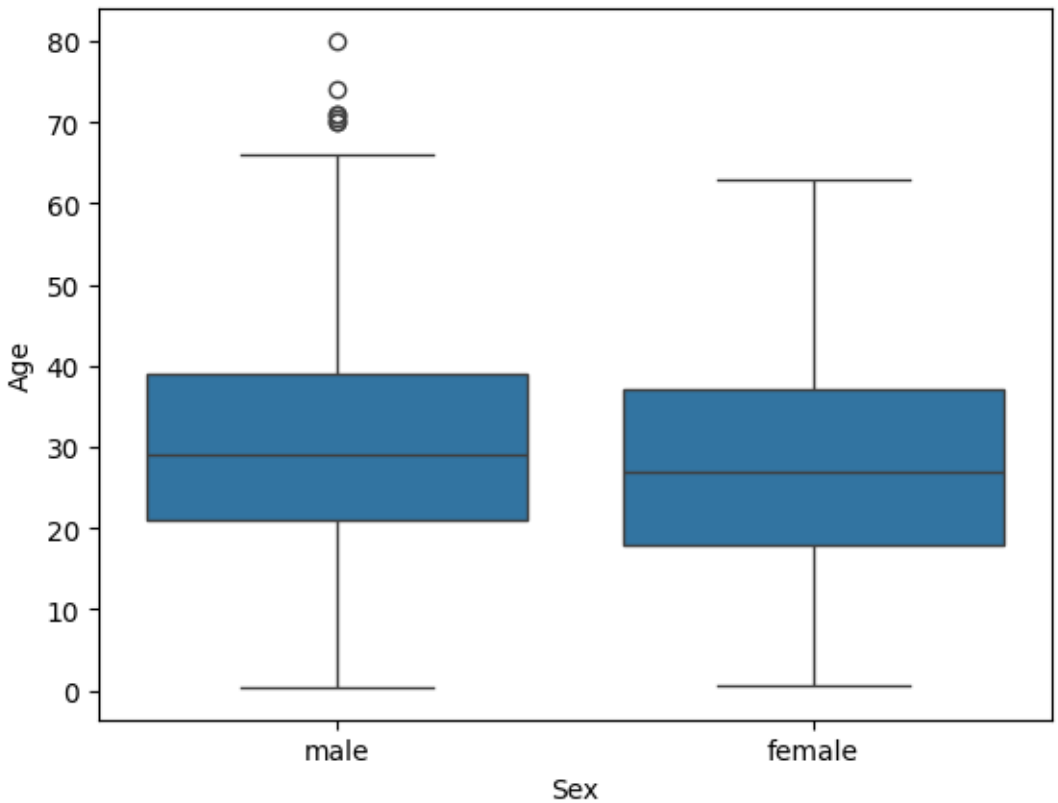
[55]:
```
## Multivariate Bar Plot of Pclass, Fare, and Sex
```

[56]:
```
sns.barplot(x="Pclass", y="Fare", hue="Sex", data=data)
plt.show()
```



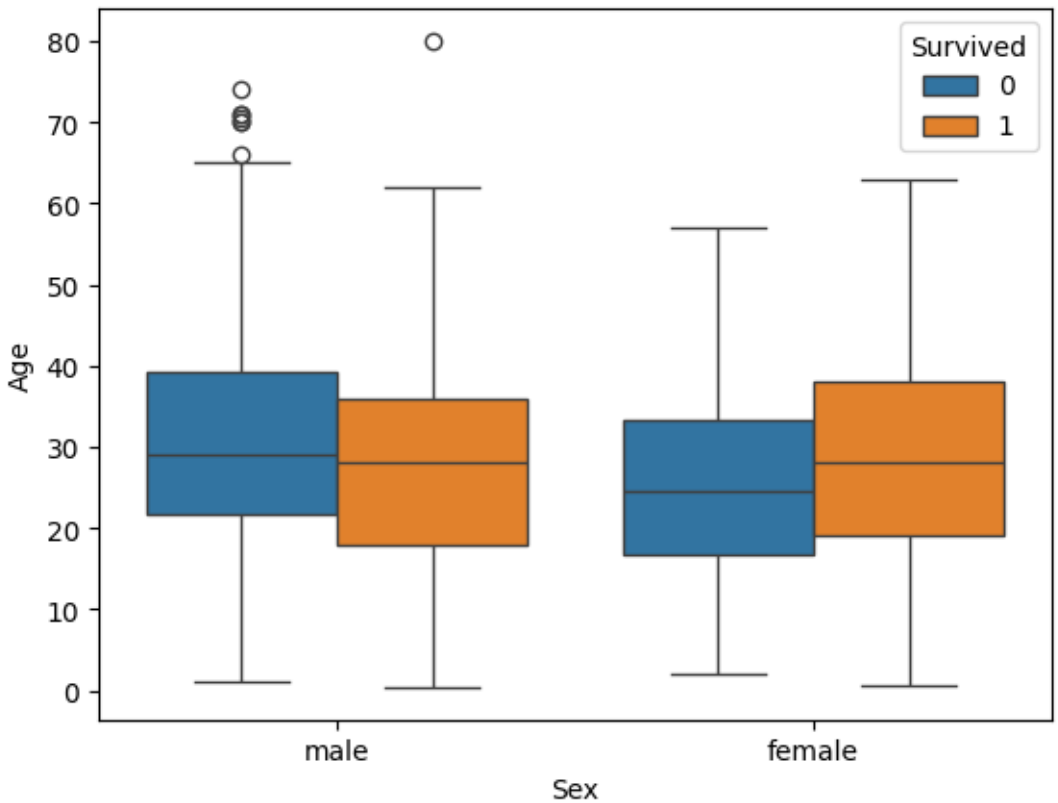
[57]:
```
## Boxplot of Sex vs Age
```

[58]:
```
sns.boxplot(x="Sex", y="Age", data=data)
plt.show()
```



[59]:
```
## Multivariate Boxplot of Sex, Age, and Survived
```

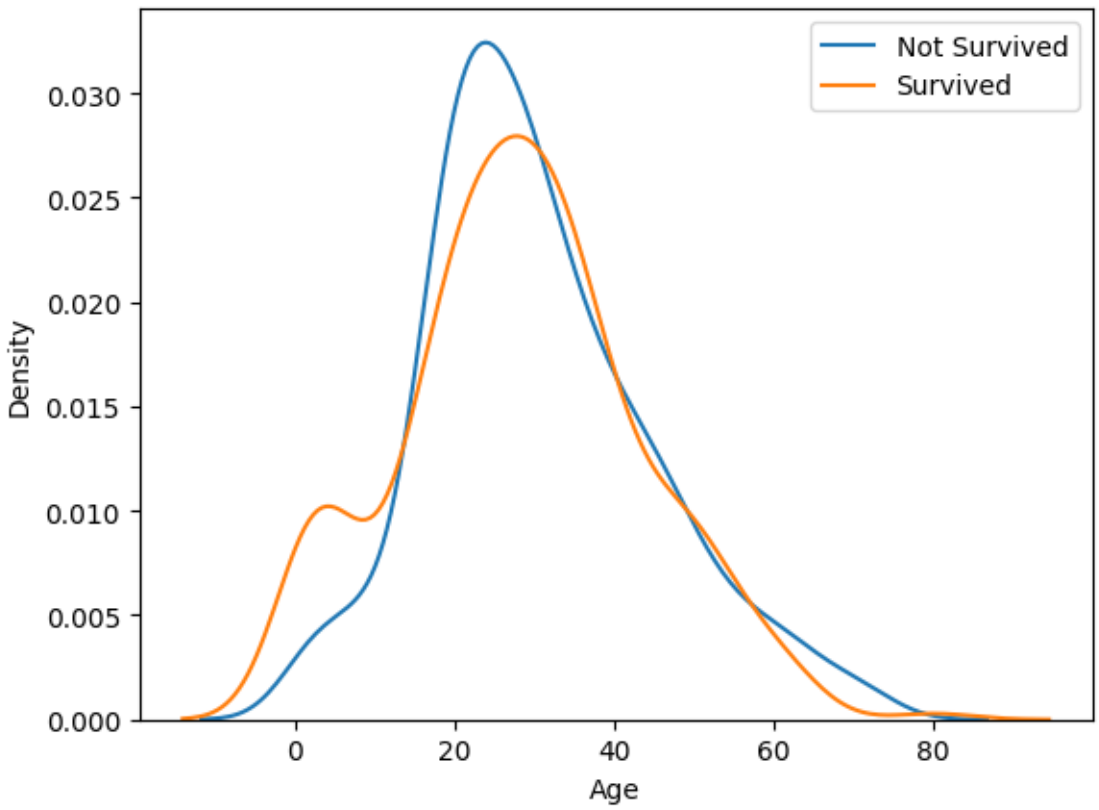
[60]:
```
sns.boxplot(x="Sex", y="Age", hue="Survived", data=data)
plt.show()
```



[61]:
```
## Distplot of Age by Survival
```

[62]:
```
sns.kdeplot(data=data[data['Survived']==0]['Age'], label="Not Survived")
sns.kdeplot(data=data[data['Survived']==1]['Age'], label="Survived")

plt.legend()
plt.show()
```



[63]:
```
## Crosstab of Pclass and Survived
```

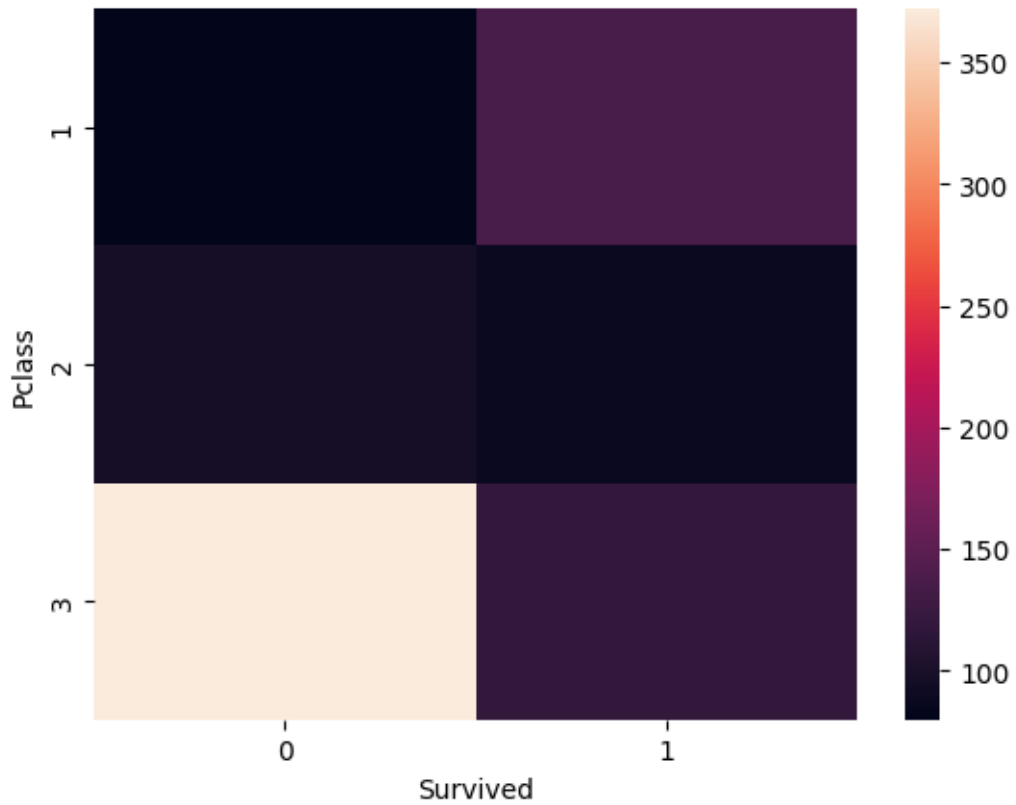
[64]:
```
pd.crosstab(data['Pclass'], data['Survived'])
```

[64]:
```
Survived    0    1
Pclass
1          80  136
2          97   87
3         372  119
```

[65]:
```
## Heatmap of Pclass and Survived
```

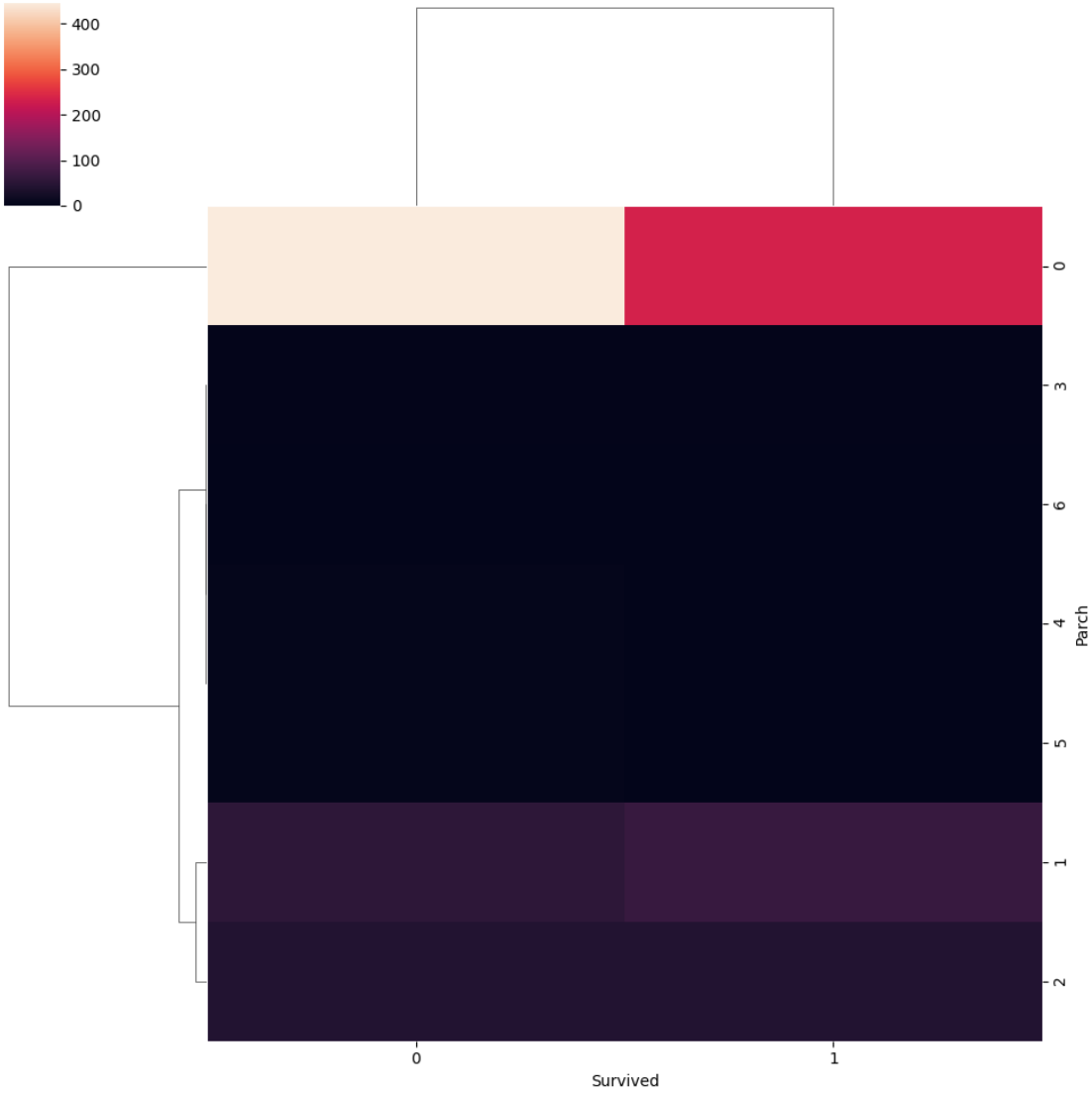
[66]:
```
sns.heatmap(pd.crosstab(data['Pclass'], data['Survived']))
```

[66]:
```
<Axes: xlabel='Survived', ylabel='Pclass'>
```



[67]:
```
## Cluster Map of Parch and Survived
```

[68]:
```
sns.clustermap(pd.crosstab(data['Parch'], data['Survived']))
plt.show()
```


400
300
200
100
0
0
3
6
4
5
1
2
Parch
0
1
Survived

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